CLASS 9 · CBSE · GANITA MANJARI PART 1 · DPP SHEET

Chapter 7: The Mathematics of Maybe: Introduction to Probability

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Write the probability of an impossible event and of a certain event.
2. A die is rolled 40 times and shows 5 on nine of them. Write the experimental and the theoretical probability of getting 5.
3. A die is rolled once. Find the probability of getting a number less than 3.
4. A letter is chosen at random from the word STATISTICS. Find the probability that it is (a) T (b) a vowel.
5. In a survey of 80 students, 28 prefer cricket. Estimate how many of the 1200 students of the school prefer cricket.
6. Write the sample space for tossing a coin and rolling a die together. How many outcomes does it have?
7. Two coins are tossed. Find the probability of getting at least one tail.
8. Three coins are tossed. Find the probability of getting exactly one head.
9. Cards numbered 1 to 20 are placed in a box and one is drawn. Find the probability that the number is (a) a multiple of 4 (b) prime.
10. A bag has 3 red and 2 green balls. A ball is drawn, put back, and a ball is drawn again. Find the probability that both are red.

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Answer Key

1. 0 and 1
2. $\frac{9}{40} = 0.225$ and $\frac{1}{6}$
3. $\frac{2}{6} = \frac{1}{3}$
4. (a) $\frac{3}{10}$ (b) $\frac{3}{10}$
5. $0.35 \times 1200 = 420$
6. $\{H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6\}$; 12 outcomes
7. $\frac{3}{4}$
8. $\frac{3}{8}$
9. (a) $\frac{5}{20} = \frac{1}{4}$ (b) $\frac{8}{20} = \frac{2}{5}$
10. $\frac{3}{5} \times \frac{3}{5} = \frac{9}{25}$

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